

Time series for the development of renewable energy sources in Germany

based on statistical data from the
Working Group on Renewable Energy-Statistics (AGEE-Stat)
(Status: September 2025)

For our environment



**Umwelt
Bundesamt** 

INTRODUCTION

DEVELOPMENT OF RENEWABLE ENERGY SOURCES IN GERMANY

Tab. 1	Development of renewable energy sources
Tab. 2	Shares of renewable energy sources
Tab. 2.1	Development of renewable energies according to international reporting conventions (SHARES Indicators)
Tab. 3	Gross electricity production from renewable energy sources
Tab. 3.1	Net electricity production from renewable energy sources
Tab. 4	Installed electrical capacity of renewable energy plants
Tab. 4.1	Additional differentiation: Solar Photovoltaik
Tab. 5	Final energy consumption from renewable sources for heating and cooling
Tab. 5.1	Renewable energy sources in the production of district heating
Tab. 5.2	Final energy consumption from renewable sources for heating and cooling by sectors
Tab. 5.3	Additional differentiation: Ground source geothermal energy and environmental heat (heat pumps)
Tab. 6	Final energy consumption from renewable sources in the transport sector

DEVELOPMENT OF TOTAL FINAL ENERGY CONSUMPTION IN GERMANY

Tab. 7	Development of energy consumption in Germany
--------	--

RENEWABLE ENERGY SOURCES AND ENVIRONMENT

Tab. 8.1	Greenhouse gas emissions avoided through the use of renewable energy sources
Tab. 8.2	Acidifying emissions avoided through the use of renewable energy sources
Tab. 8.3	Avoidance factors of renewable energy sources

ECONOMIC STIMULI FROM RENEWABLE ENERGY SOURCES

Tab. 9.1	Investments in construction of renewable energy plants
Tab. 9.2	Economic stimuli from the operation of renewable energy plants

RENEWABLE ENERGY SOURCES IN YEARS

2023-2024	Detailed data on renewable energy sources for the years 2023/2024
-----------	---

ANNEX: CHANGES IN METHODOLOGY, CONVERSION FACTORS AND ABBREVIATIONS

LIST OF INSTITUTIONS, REFERENCES AND LITERATURE

[back to directory](#)

The Working Group on Renewable Energy Statistics (AGEE-Stat) was established in February 2004 in order to place statistics and data relating to renewable energy sources on a comprehensive, up-to-date and coordinated basis (more information on: www.erneuerbare-energien.de/EE/ee-in-zahlen-arbeitsgruppe).

With the increasing share of renewable energy sources in the energy system and increasing national and international reporting obligations the demand on reliable data concerning the development of all renewable energy sources in Germany has been growing. Therefore, the coordination office of the Working Group engages in various research projects to improve the data basis and the scientific calculation methods- supported by the working group members (ministries, federal offices and academic institutions).

To analyze the historic development of the renewable energy sources in Germany since 1990 several data sources were evaluated and supplemented by (model-based) calculations. Wherever possible official data were used. Partially missing values were extrapolated from existing data. Single data gaps were closed by estimations, which is why the data are governed by a certain degree of uncertainties.

In some cases data are not available before a specific point of time and previous years could not be estimated reliable, but if new information allow conclusions to be drawn on historic values these data will be complemented or corrected. In addition, changes of the data collection methods or categories may result in breaks in time series, but only in cases in which a retrospective amendment is not possible and reasonable. Rounding may also cause discrepancies in the totals. Therefore the published data are partially preliminary and can differ in comparison to previous issues.

Detailed background information on data sources and methodology of the time series of the gross electricity production and installed electrical capacity can be found on: www.umweltbundesamt.de/publikationen/datenquellen-methodik-der-agee-stat-zeitreihen-zur (available in German only).

Beside these time tables the AGEE-Stat publishes monthly and quarterly reports.

These can be found on:

www.umweltbundesamt.de/themen/klima-energie/erneuerbare-energien/erneuerbare-energien-in-zahlen/monats-quartalsberichte-der-agee-stat

Further enquiries to the data and methodology can be addressed on

Federal Environment Agency
Section V 1.8 - Monitoring of Renewable Energies,
Coordination office of the Working Group on Renewable Energy Statistics (AGEE-Stat)
Wörlitzer Platz 1
06844 Dessau-Roßlau
E-Mail: AGEE-stat@uba.de

Table 1: Development of renewable energy sources

values before 2005 hidden

Status: September 2025

[back to directory](#)

Figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Gross final energy consumption ¹⁾	188.019	227.802	261.777	267.712	264.252	305.863	316.086	351.782	364.416	358.673	387.587	389.338	420.841	434.525	456.289	472.078	472.059	497.438	502.014	514.305
Gross electricity production	63.538	72.795	89.999	95.106	96.974	106.446	125.648	145.091	153.677	163.740	190.073	191.107	217.674	225.291	243.590	253.543	239.684	255.600	275.557	285.904
Production of district heating	6.892	7.899	8.261	9.744	10.575	11.565	12.640	14.885	16.035	16.428	17.989	18.528	18.791	21.347	22.130	22.346	24.264	24.095	24.305	25.414
Final energy consumption for heating and cooling (including district heat) ²⁾	102.588	115.307	126.850	137.227	135.823	164.904	156.310	170.760	177.734	160.697	165.566	165.992	170.521	174.885	179.081	176.943	195.145	204.110	186.757	191.505
Final energy consumption for heating and cooling (without electricity and district heat) ³⁾	96.315	108.136	119.311	128.335	126.187	154.307	144.931	157.445	163.444	146.326	149.752	149.793	154.082	156.474	159.426	157.122	173.628	182.891	165.421	169.181
thereof in: Households	57.329	67.465	71.229	82.240	75.208	90.993	84.599	97.371	101.640	85.863	87.620	85.578	88.687	91.491	93.127	90.015	101.992	112.318	94.760	96.693
thereof in: TCS	14.441	16.806	19.515	24.092	26.432	31.861	26.589	29.197	31.174	28.865	31.670	31.952	33.323	33.552	34.983	36.030	38.812	37.090	38.515	39.472
thereof in: Industry	24.545	23.865	28.567	22.003	24.547	31.453	33.743	30.877	30.630	31.598	30.462	32.263	32.072	31.431	31.316	31.077	32.823	33.483	32.145	33.016
Final energy consumption in the transport sector ⁴⁾	22.627	40.456	45.968	36.226	32.441	35.622	35.361	37.223	34.277	35.348	33.296	33.642	34.623	35.994	36.041	44.304	39.850	41.376	44.834	42.977

Figures in [PJ]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Primary energy consumption ⁵⁾	788	962	1.116	1.156	1.154	1.317	1.369	1.534	1.584	1.576	1.682	1.687	1.798	1.832	1.909	1.974	1.970	2.050	2.054	2.094

differences in the totals may occur due to roundings

- 1) according to the German government's Energy Concept
- 2) sectoral local consumption of energy for application purposes of heating and cooling, without electricity, including district heat
- 3) sectoral local consumption of energy for application purposes of heating and cooling, without secondary energy carriers electricity and district heat
- 4) consumption of biogenic fuels and renewables-based electricity in the transport sector (excluding consumption in agriculture, forestry and military)
- 5) till 2002 by Working Group on Energy Balances (AGEB), calculated according to the "physical energy content" principle, since 2003 by AGEE-Stat based on JAQ-REN

Sources: AGEE-Stat based on AGEB [1]; Eurostat [18]; IEA [22]; EP/ER [16], [17] and further sources see table 3, 5 und 6; partially preliminary data

Table 2: Shares of renewable energy sources

values before 2005 hidden

Status: September 2025

[back to directory](#)

Figures in [%]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
of gross national electricity consumption	10,3	11,7	14,4	15,3	16,6	17,2	20,6	23,8	25,3	27,5	31,6	31,8	36,2	37,9	42,2	45,4	41,7	46,3	52,9	54,1
of district heat production ¹⁾	5,1	5,8	6,3	7,3	8,1	8,1	9,7	11,1	11,8	13,5	14,2	14,2	14,4	16,4	17,4	18,9	17,6	20,1	21,1	21,1
of final energy consumption for heating and cooling (including district heat) ²⁾	8,0	8,6	10,6	10,4	11,2	12,2	12,6	13,6	13,6	13,7	13,6	13,4	13,7	14,5	14,6	14,7	15,4	17,5	17,4	17,8
of final energy consumption for heating and cooling (without secondary energy carriers electricity and district heat) ³⁾	8,4	8,9	11,0	10,8	11,6	12,7	12,9	13,8	13,8	13,7	13,5	13,3	13,7	14,3	14,3	14,3	15,2	17,2	17,0	17,4
thereof in: Households	10,7	12,2	15,7	15,2	14,9	16,5	18,1	19,1	18,9	19,1	18,6	17,6	18,5	18,9	18,2	17,8	19,9	23,1	21,5	22,1
thereof in: TCS	7,5	7,3	10,8	11,6	13,5	15,2	13,9	15,5	14,7	15,0	15,9	16,8	17,3	19,8	20,3	20,9	21,9	22,5	25,4	26,4
thereof in: Industry	5,8	5,5	6,4	5,0	6,3	6,9	7,3	7,0	7,0	7,4	7,0	7,2	7,1	7,1	7,3	7,5	7,2	8,1	8,4	8,6
of final energy consumption in the transport sector	3,6	6,2	7,4	5,9	5,4	5,8	5,7	6,1	5,5	5,6	5,2	5,2	5,2	5,6	5,5	7,5	6,7	6,9	7,6	7,3

Figures in [%]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
of primary energy consumption	5,4	6,5	7,9	8,0	8,5	9,3	10,1	11,3	11,4	11,9	12,6	12,5	13,3	13,9	14,9	16,6	15,8	17,5	19,3	19,9

1) Net heat production, including grid losses
3) sectoral local consumption of energy for application purposes of heating and cooling, without electricity, including district heat
4) sectoral local consumption of energy for application purposes of heating and cooling, without secondary energy carriers electricity and district heat

Sources: see table 1 and 7; partially preliminary data

Table 2.1: Development of renewable energies according to international reporting conventions (SHARES Indicators)

values before 2005 hidden

Status: February 2025

[back to directory](#)

RED	Period of Validity [in Years]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	Reporting based on	RED I																RED II			
RES	Gross final energy consumption	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	Renewable Energies [TWh]	188,8	228,7	256,7	268,1	271,0	311,7	316,6	350,7	364,7	362,1	382,3	388,6	407,4	432,5	446,4	465,4	484,6	512,3	502,8	519,3
	Total [TWh]	2.635,0	2.701,1	2.557,2	2.661,4	2.497,5	2.671,4	2.538,9	2.588,3	2.650,8	2.517,5	2.565,0	2.609,7	2.632,5	2.595,8	2.585,6	2.437,8	2.516,7	2.451,1	2.331,9	2.315,0
	Share of renewable energy sources [%]	7,2	8,5	10,0	10,1	10,9	11,7	12,5	13,5	13,8	14,4	14,9	14,9	15,5	16,7	17,3	19,1	19,3	20,9	21,6	22,4
RESE	Gross electricity consumption	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	Renewable Energies [TWh]	68,4	75,4	85,7	91,3	99,0	109,6	125,4	141,3	151,1	165,0	184,4	193,8	208,2	222,1	232,5	244,4	249,6	261,3	269,4	284,4
	Total [TWh]	611,1	615,8	617,2	614,6	578,2	611,3	603,1	601,1	600,0	587,7	593,6	593,8	594,8	585,1	567,8	549,4	568,9	545,7	515,6	517,6
	Share of renewable energy sources [%]	11,2	12,2	13,9	14,9	17,1	17,9	20,8	23,5	25,2	28,1	31,1	32,6	35,0	38,0	41,0	44,5	43,9	47,9	52,2	54,9
RES-H&C	Final energy consumption for heating and cooling (including district heat)	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	Renewable Energies [TWh]	99,7	112,0	121,8	133,0	130,6	157,3	148,9	165,1	171,0	152,9	156,8	154,0	157,4	165,6	167,6	163,6	204,1	217,1	197,7	202,4
	Total [TWh]	1.332,2	1.378,9	1.243,5	1.354,8	1.241,6	1.374,3	1.246,4	1.293,9	1.343,9	1.216,0	1.248,3	1.271,5	1.271,5	1.260,3	1.256,2	1.246,3	1.314,2	1.230,6	1.155,9	1.139,9
	Share of renewable energy sources [%]	7,7	8,4	10,2	10,3	11,2	12,1	12,6	13,4	13,4	13,4	13,4	13,0	13,4	14,2	14,5	14,5	15,5	17,4	17,1	17,8
RES-H&C DH	District heating	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	Production of Renewable Energies [TWh]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	24,2	24,0	23,9	-
	Final energy consumption for district heat [TWh]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	130,1	119,5	114,1	-
	Share of renewable energy sources [%]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	18,6	20,1	20,9	-
RES-T	Final energy consumption in the transport sector	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	Renewable Energies [TWh]	24,4	41,9	45,9	37,9	34,6	38,3	39,0	43,9	44,7	42,9	41,4	45,2	46,1	50,3	48,9	59,1	47,0	58,9	69,8	68,2
	Total [TWh]	616,4	625,0	609,4	599,7	588,9	596,9	604,4	600,0	612,8	621,4	630,6	644,4	656,4	633,9	640,8	590,3	580,3	594,2	588,1	586,4
	Share of renewable energy sources [%]	4,0	6,7	7,5	6,3	5,9	6,4	6,5	7,3	7,3	6,9	6,6	7,0	7,0	7,9	7,6	10,0	8,1	9,9	11,9	11,6

Table 3: Gross electricity production from renewable energy sources

values before 2005 hidden

Status: September 2025

[back to directory](#)

Figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Hydropower ¹⁾	19.638	20.031	21.170	20.443	19.031	20.953	17.671	21.755	22.998	19.587	18.977	20.546	20.150	18.098	20.135	18.721	19.657	17.625	19.895	22.477
Wind energy onshore	27.774	31.324	40.507	41.385	39.382	38.371	49.280	50.948	51.819	57.026	72.340	67.650	88.018	90.484	101.150	104.796	92.075	99.149	117.877	112.846
Wind energy offshore	0	0	0	0	38	176	577	732	918	1.471	8.284	12.274	17.675	19.467	24.744	27.306	24.373	25.124	23.887	26.068
Solar Photovoltaic	1.308	2.265	3.137	4.508	6.715	11.963	19.991	26.744	30.621	35.448	38.076	37.556	38.761	44.330	45.212	49.500	50.486	61.287	64.300	75.379
Solid biofuels ²⁾	7.478	8.819	8.699	9.296	9.746	10.351	10.517	10.694	10.555	10.797	11.034	10.798	10.644	11.068	11.038	11.306	10.738	10.663	10.460	10.147
Liquid biofuels	121	749	987	1.133	1.700	1.331	398	355	299	347	443	508	454	397	344	320	210	97	110	96
Biogas	1.803	3.557	8.915	11.648	14.020	16.265	19.936	25.920	27.468	28.614	30.086	30.726	31.089	30.266	30.106	30.671	31.011	30.970	28.272	28.641
Biomethane	0	0	21	46	81	385	596	1.117	1.713	2.512	3.147	3.146	2.966	2.980	2.990	3.046	3.273	3.098	3.120	2.796
Sewage gas	1.096	1.057	1.033	1.094	1.131	1.203	1.280	1.314	1.308	1.336	1.389	1.440	1.460	1.555	1.581	1.579	1.576	1.547	1.529	1.529
Landfill gas	1.068	1.092	1.009	864	788	674	628	536	483	435	396	358	338	305	287	247	249	206	211	177
Biogenic fraction of waste ³⁾	3.252	3.901	4.521	4.671	4.323	4.746	4.755	4.951	5.415	6.069	5.768	5.930	5.956	6.163	5.806	5.820	5.792	5.628	5.701	5.533
Geothermal energy	0,2	0,4	0,4	18	19	28	19	25	80	98	133	175	163	178	197	231	244	206	195	215
Total	63.538	72.795	89.999	95.106	96.974	106.446	125.648	145.091	153.677	163.740	190.073	191.107	217.674	225.291	243.590	253.543	239.684	255.600	275.557	285.904
for information: amount of electricity under the EEG ⁴⁾	43.967	51.545	67.010	71.148	75.377	82.332	103.136	118.330	125.693	136.936	162.730	162.348	188.300	196.306	212.765	221.956	203.362	219.765	235.565	238.062

1) river and storage power plants including pumped storage plants with natural inflow

2) including sewage sludge

3) biogenic fraction of waste in waste incineration plants estimated at 50 %, from 2008 only municipal waste

4) fixed remuneration, market premium (since 2012), "green electricity privilege" and other direct marketing as well as the amount of remunerated self consumed electricity of photovoltaic plants under the EEG regulations between 2009-2012, including mine gas

annual statement of account for the EEG of the transmission system operators: www.netztransparenz.de

Sources: AGEE-Stat based on StBA [26], [27]; BNetzA [8]; ÜNB; ZSW; DENA [24]; BDEW; VDEW [20]; AGEb [1]; DBFZ [13]; IE [21]; partially preliminary data

Table 3.1: Net electricity production from renewable energy sources

values before 2005 hidden

Status: September 2025

[back to directory](#)

Figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Hydropower ¹⁾	19.344	19.718	20.811	20.134	18.743	20.682	17.326	21.332	22.660	19.310	18.665	20.215	19.985	17.926	19.957	18.546	19.483	17.458	19.722	22.276
Wind energy onshore	27.229	30.710	39.713	40.574	38.610	37.619	48.314	49.949	50.803	55.908	70.922	66.324	86.293	88.710	99.166	102.741	90.270	97.205	115.565	110.634
Wind energy offshore	0	0	0	0	38	174	568	722	905	1.449	8.162	12.092	17.414	19.179	24.379	26.903	24.013	24.752	23.534	25.682
Solar Photovoltaic	1.282	2.220	3.075	4.420	6.583	11.729	19.599	26.220	30.020	34.753	37.330	36.820	38.001	43.461	44.326	48.529	49.496	60.085	63.039	73.901
Solid biofuels ²⁾	6.721	7.913	7.824	8.285	8.733	9.247	9.395	9.509	9.392	9.657	9.855	9.673	9.542	9.807	9.790	10.002	9.594	9.525	9.385	9.109
Liquid biofuels	117	726	958	1.099	1.649	1.291	386	319	286	333	425	481	428	381	330	308	201	91	104	95
Biogas	1.749	3.450	8.647	11.298	13.599	15.777	19.338	25.143	26.644	27.756	29.184	29.804	30.156	29.358	29.206	29.754	30.084	30.044	27.427	27.785
Biomethane	0	0	20	44	79	376	582	1.091	1.672	2.452	3.071	3.070	2.895	2.930	2.923	2.977	3.195	3.024	3.045	2.729
Sewage gas	1.057	1.023	999	1.033	1.042	1.093	1.179	1.217	1.215	1.283	1.308	1.331	1.357	1.551	1.578	1.576	1.572	1.544	1.526	1.527
Landfill gas	1.040	1.046	962	826	713	600	595	512	461	416	384	343	324	279	262	233	234	193	198	165
Biogenic fraction of waste ³⁾	2.422	2.965	3.519	3.667	3.352	3.755	3.795	3.971	4.305	4.838	4.565	4.746	4.802	4.932	4.626	4.638	4.575	4.436	4.521	4.340
Geothermal energy	0	0	0	12	13	20	19	25	69	67	91	164	157	126	144	173	174	152	139	148
Total	60.961	69.771	86.528	91.392	93.154	102.363	121.096	140.010	148.432	158.222	183.962	185.063	211.354	218.640	236.687	246.380	232.891	248.509	268.205	278.391

1) river and storage power plants including pumped storage plants with natural inflow
2) including sewage sludge
3) biogenic fraction of waste in waste incineration plants estimated at 50 %, from 2008 only municipal waste
annual statement of account for the EEG of the transmission system operators: www.netztransparenz.de

Sources: AGEE-Stat based on StBA [26], [27]; BNetzA [8]; ÜNB; ZSW; DENA [24]; BDEW; VDEW [20]; AGEb [1]; DBFZ [13]; IE [21]; partially preliminary data

Table 4: Installed electrical capacity of renewable energy plants

values before 2005 hidden

Status: September 2025

[back to directory](#)

Figures in [MW]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Hydropower ¹⁾	5.210	5.193	5.137	5.164	5.340	5.407	5.625	5.607	5.590	5.580	5.589	5.629	5.627	5.347	5.396	5.454	5.489	5.569	5.607	5.500
Wind energy onshore	18.248	20.474	22.116	22.794	25.697	26.823	28.524	30.711	32.969	37.620	41.297	45.283	50.174	52.328	53.187	54.239	55.870	57.975	60.965	63.530
Wind energy offshore ²⁾	0	0	0	0	35	80	188	268	508	994	3.283	4.152	5.406	6.393	7.555	7.874	7.874	8.216	8.473	9.215
Solar Photovoltaic	2.056	2.897	4.171	6.120	10.566	18.005	25.917	34.079	36.710	37.899	39.225	40.679	42.293	45.205	48.862	54.464	60.198	67.945	83.995	102.062
Solid biofuels	1.218	1.411	1.431	1.457	1.470	1.502	1.554	1.558	1.623	1.589	1.592	1.600	1.601	1.567	1.568	1.571	1.587	1.592	1.590	1.615
Liquid biofuels	60	177	295	341	412	410	345	277	263	232	232	231	230	230	231	231	201	196	192	187
Biogas ³⁾	665	1.000	1.226	1.419	2.520	3.015	3.837	4.212	4.317	4.380	4.601	4.780	5.173	5.610	5.951	6.343	6.453	6.516	6.575	6.612
Biomethane ³⁾	0	0	6	16	18	96	218	256	383	603	614	653	567	602	603	621	629	630	641	690
Sewage gas ⁴⁾	161	170	177	186	192	200	233	236	240	245	245	245	242	421	396	374	373	371	368	369
Landfill gas	248	252	257	268	261	237	232	214	210	211	183	172	165	169	162	156	137	132	131	129
Geothermal energy	0	0	3	3	4	4	4	15	25	28	28	34	34	34	39	39	46	51	50	50
Biogenic fraction of waste ⁵⁾	587	637	614	684	720	762	743	714	930	944	962	978	1.004	1.063	1.084	1.024	1.040	1.023	1.004	1.023
Total	28.453	32.211	35.433	38.452	47.235	56.541	67.420	78.147	83.768	90.325	97.851	104.436	112.516	118.969	125.034	132.390	139.897	150.216	169.591	190.982

1) river and storage power plants including pumped storage plants with natural inflow

2) installed capacity of offshore wind energy plants connected to the network

3) since 2013 including additional capacity for increased flexibility of electricity production

4) till 2014 calculated based on the electricity production and full-load hours specific to the energy source, since 2015 net additions according to registry data of the Federal Network Agency (BNetzA), since 2018 based on StBA

5) the stated values represent an amount of 50% of the total installed capacity of thermal waste combustion plants using municipal renewable and fossil waste. For the whole time series 50% of the total capacity is considered as renewable.

Sources: AGEE-Stat based on BNetzA [8]; StBA [27]; ZSW, DENA [24]; BDEW; VDEW [20], DBFZ [13], DEWI [14]; IE [21]; partially preliminary data

values before 2005 hidden
[back to directory](#)

Status: September 2025

gross electrical capacity	figures in [MW]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	building-mounted PV systems	1.869	2.637	3.714	5.414	8.992	14.709	20.110	24.530	26.134	26.899	27.603	28.467	29.554	31.529	34.287	38.148	41.662	46.302	57.918	69.422
	thereof: ≤ 2 kW ¹⁾	32	35	38	40	42	44	47	49	51	52	53	54	55	56	58	68	104	348	1.239	2.524
	thereof: 2 < x < 10 kW	590	811	1.067	1.440	1.959	2.644	3.444	4.206	4.666	4.896	5.127	5.392	5.745	6.148	6.718	7.961	9.040	10.614	13.958	16.070
	thereof: 10 ≤ x < 20 kW	403	565	782	1.117	1.729	2.593	3.461	4.042	4.255	4.329	4.371	4.405	4.445	4.498	4.576	4.735	5.424	6.437	9.965	13.293
	thereof: 20 ≤ x < 30 kW	372	532	772	1.134	1.889	2.957	3.742	4.265	4.421	4.485	4.530	4.581	4.664	4.778	4.954	5.257	5.617	6.008	6.981	8.109
	thereof: 30 ≤ x < 100 kW	343	488	727	1.144	2.128	3.742	5.079	6.012	6.304	6.446	6.546	6.688	6.903	7.252	7.698	8.258	8.709	9.139	10.004	11.050
	thereof: 100 ≤ x < 500 kW	100	154	246	399	911	2.049	3.237	4.409	4.752	4.936	5.182	5.288	5.516	6.049	6.843	7.636	8.153	8.619	9.632	10.865
	thereof: 500 ≤ x ≤ 750 kW	6	13	22	37	109	269	453	630	684	708	717	753	914	1.428	2.119	2.884	3.210	3.421	3.762	4.144
	thereof: 700 < x ≤ 1.000 kW	4	9	12	29	105	214	349	475	499	519	535	634	635	636	637	648	657	695	908	1.244
	thereof > 1.000 kW	19	30	48	74	120	197	298	442	502	528	542	672	677	684	684	701	748	1.021	1.469	2.114
	figures in [MW]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	ground-mounted PV systems	187	260	457	706	1.574	3.296	5.807	9.549	10.576	11.000	11.622	12.212	12.739	13.676	14.575	16.316	18.536	21.643	26.077	32.640
	thereof: < 10 kW	13	17	20	23	28	37	45	53	58	60	63	66	70	74	80	89	98	111	143	156
	thereof: 10 ≤ x < 30 kW	16	21	31	44	66	99	124	140	144	146	148	149	151	154	157	161	168	176	205	221
	thereof: 30 ≤ x < 100 kW	31	37	47	62	102	165	215	251	260	265	268	270	278	288	299	313	324	334	355	386
	thereof: 100 ≤ x ≤ 750 kW	33	46	65	95	168	307	430	629	692	721	750	761	865	1.223	1.570	2.036	2.292	2.413	2.536	2.708
	thereof: 750 < x ≤ 1.000 kW	7	10	14	29	71	130	197	354	405	425	448	462	468	471	479	501	553	588	716	992
	thereof: 1.000 < x ≤ 10.000 kW	87	129	280	443	1.032	2.241	4.104	6.343	7.238	7.604	8.166	8.725	9.128	9.666	10.178	11.082	12.312	14.179	15.970	18.272
	thereof > 10.000 kW	0	0	0	10	107	317	692	1.779	1.779	1.779	1.779	1.779	1.779	1.800	1.812	2.134	2.789	3.842	6.152	9.905
	Total	2.056	2.897	4.171	6.120	10.566	18.005	25.917	34.079	36.710	37.899	39.225	40.679	42.293	45.205	48.862	54.464	60.198	67.945	83.995	102.062

net electrical capacity	Figures in [MW]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	building-mounted PV systems	1.662	2.353	3.325	4.869	8.165	13.457	18.488	22.642	24.136	24.840	25.481	26.265	27.254	29.032	31.523	34.998	38.150	42.283	52.507	62.670
	thereof: ≤ 2 kW ¹⁾	30	33	35	37	39	41	44	46	47	48	49	50	51	52	54	64	97	308	1.004	2.124
	thereof: 2 < x < 10 kW	529	728	958	1.294	1.769	2.401	3.149	3.867	4.296	4.509	4.721	4.964	5.287	5.657	6.184	7.335	8.329	9.784	12.896	14.870
	thereof: 10 ≤ x < 20 kW	353	497	690	990	1.547	2.340	3.154	3.701	3.900	3.968	4.006	4.038	4.075	4.124	4.196	4.340	4.947	5.822	8.855	11.728
	thereof: 20 ≤ x < 30 kW	328	472	689	1.020	1.720	2.711	3.445	3.938	4.084	4.144	4.185	4.233	4.309	4.415	4.578	4.855	5.181	5.530	6.400	7.412
	thereof: 30 ≤ x < 100 kW	304	435	652	1.033	1.940	3.435	4.674	5.551	5.824	5.956	6.047	6.177	6.375	6.696	7.105	7.618	8.031	8.425	9.212	10.165
	thereof: 100 ≤ x < 500 kW	91	140	225	367	841	1.899	3.004	4.106	4.425	4.593	4.815	4.911	5.116	5.590	6.295	6.985	7.433	7.836	8.708	9.776
	thereof: 500 ≤ x ≤ 750 kW	6	13	21	34	101	250	420	586	635	656	665	697	839	1.289	1.901	2.568	2.849	3.033	3.330	3.662
	thereof: 700 < x ≤ 1.000 kW	4	8	11	27	98	199	326	444	467	485	500	589	590	591	592	601	609	641	823	1.113
	thereof > 1.000 kW	17	27	44	67	110	181	272	403	458	481	493	606	612	618	618	632	674	904	1.279	1.820
	Figures in [MW]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	ground-mounted PV systems	169	240	418	646	1.447	3.048	5.344	8.761	9.677	10.050	10.600	11.116	11.568	12.384	13.164	14.679	16.579	19.249	23.012	28.534
	thereof: < 10 kW	13	16	18	22	25	34	42	49	54	56	58	61	65	69	75	83	91	103	133	145
	thereof: 10 ≤ x < 30 kW	14	19	28	40	60	90	114	129	133	135	136	137	140	142	145	149	154	162	188	203
	thereof: 30 ≤ x < 100 kW	27	33	43	57	93	152	198	232	241	246	249	251	257	267	277	290	301	310	329	357
	thereof: 100 ≤ x ≤ 750 kW	30	43	60	88	156	286	402	590	649	675	702	713	805	1.123	1.434	1.851	2.078	2.183	2.290	2.441
	thereof: 750 < x ≤ 1.000 kW	6	10	13	27	66	121	184	332	380	399	419	431	437	440	446	466	511	542	654	897
	thereof: 1.000 < x ≤ 10.000 kW	79	119	256	403	942	2.070	3.775	5.830	6.621	6.940	7.437	7.924	8.265	8.726	9.160	9.939	10.990	12.588	14.117	16.088
	thereof > 10.000 kW	0	0	0	9	105	295	629	1.599	1.599	1.599	1.599	1.599	1.599	1.617	1.627	1.901	2.454	3.361	5.301	8.403
	Total	1.831	2.593	3.743	5.515	9.612	16.505	23.832	31.403	33.813	34.890	36.081	37.381	38.822	41.416	44.687	49.677	54.729	61.532	75.519	91.204

1) The value includes, in addition to the 0.8 million systems reported in the German Market Master Data Register (MaSiR), an estimate from a research project of a further 2.0 million unregistered plug-in PV systems (commonly referred to as "balcony power plants"). The total installed gross capacity of all balcony power plants amounts to 2.4 GW, of which 0.8 GW are registered in the MaSiR. The total net capacity (inverter capacity) of all balcony power plants amounts to 2.0 GW, of which 0.5 GW are registered in the MaSiR.

Table 5: Final energy consumption from renewable sources for heating and cooling

values before 2005 hidden
[back to directory](#)

Status: September 2025

Figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Solid biofuels (households) ¹⁾	52.222	61.496	64.092	74.020	65.220	79.304	71.904	83.015	85.941	69.484	69.477	66.289	68.393	70.192	71.154	66.874	76.874	85.440	66.187	66.088
Solid biofuels (TCS sector) ²⁾	9.695	11.120	11.443	16.046	16.476	21.236	15.813	17.437	18.696	14.945	17.137	16.731	17.450	17.812	19.219	19.101	21.908	19.926	21.429	22.663
Solid biofuels (industry) ³⁾	21.266	20.319	22.367	20.156	22.972	28.088	29.089	27.793	25.600	26.530	25.108	27.031	26.326	24.522	23.784	23.279	24.820	25.589	25.291	25.950
Solid biofuels (HP/CHP) ⁴⁾	2.043	2.104	2.225	2.759	3.581	4.057	4.665	5.776	5.532	5.465	5.957	6.259	6.193	5.740	6.121	6.296	6.796	6.470	6.685	7.010
Liquid biofuels ⁵⁾	1.225	1.814	2.869	3.442	3.735	3.442	2.603	2.204	2.196	2.372	2.189	2.188	2.194	2.298	2.383	3.217	2.591	2.471	2.532	2.166
Biogas	875	1.495	3.935	3.728	5.412	7.811	9.352	8.903	9.458	10.639	11.588	12.306	12.980	13.304	13.783	14.339	15.015	15.857	14.906	15.432
Biomethane	0	0	23	68	136	505	760	1.329	2.533	3.268	3.970	4.039	3.740	3.861	3.983	4.216	4.982	4.797	4.982	4.768
Sewage gas	2.082	1.852	1.858	1.972	1.977	1.999	2.059	2.017	1.805	1.801	2.001	2.050	2.141	2.500	2.402	2.378	2.367	2.322	2.311	2.330
Landfill gas	231	227	210	154	155	117	101	94	93	98	120	116	107	110	107	95	82	72	65	62
Gaseous biofuels	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Biogenic fraction of waste ⁶⁾	7.199	8.433	10.747	6.662	6.530	7.260	8.140	9.033	11.645	11.380	11.807	11.669	12.669	14.508	15.308	15.060	15.601	15.033	14.278	14.535
Solarthermal energy	2.857	3.363	3.746	4.293	5.061	5.383	6.160	6.416	6.500	7.025	7.562	7.604	7.834	8.955	8.667	9.013	8.549	9.730	9.121	8.795
Geothermal energy ⁷⁾	532	525	524	550	623	689	722	805	864	1.052	969	1.146	1.168	1.308	1.369	1.427	1.575	1.822	1.796	1.915
Near-surface geothermal energy, ambient heat ⁸⁾	2.361	2.559	2.811	3.377	3.945	5.013	4.942	5.938	6.871	6.638	7.681	8.564	9.326	9.775	10.801	11.648	13.985	14.581	17.174	19.791
Total	102.588	115.307	126.850	137.227	135.823	164.904	156.310	170.760	177.734	160.697	165.566	165.992	170.521	174.885	179.081	176.943	195.145	204.110	186.757	191.505

1) till 2004 according to the Working Group on Energy Balances (AGEB); since 2005 according to Thünen Institute; including charcoal
2) TCS = trade, commerce and service sector; Final energy consumption for heat only production according to Thünen Institute plus fuel input for heat production in decentralised CHP plants; including charcoal; since 2018 including sewage sludge
3) in accordance with Section 8 Energy Statistics Act; including sewage sludge
4) in accordance with Section 3 and 5 Energy Statistics Act, including sewage sludge; HP = heating plants, CHP plant = combined heat and power plant
5) including consumption of biodiesel in agriculture, forestry and military; since 2010 including blended bioethanol
6) biogenic fraction of waste in waste incineration plants estimated at 50 %, since 2008 municipal waste only, decrease 2008 due to first-time inclusion of newly available data (statistical adjustment)
7) before 2003 balneological plants are not taken into account
8) Based on GZB and FH-ISE, renewable heat from heat pumps (air-water, water-water, brine-water, process water and gas heat pumps)

Sources: AGEE-Stat based on StBA [26], [28]; ZSW; AGEb [1]; TI [23], [29], [31]; FNR [19]; Uni HH [15]; DENA [24]; DBFZ; LIAG; GZB [2]; BDH; BSW; DEPV; BWP; partially preliminary data

Table 5.1: Renewable energy sources in the production of district heating

values before 2005 hidden

Status: September 2025

[back to directory](#)

Figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Biogenic fraction of waste ¹⁾	4.378	5.304	5.482	6.373	6.105	6.563	6.728	7.414	8.371	8.225	8.412	8.518	8.993	10.010	10.133	9.628	10.541	10.076	10.295	10.790
Solid biofuels ²⁾	2.242	2.314	2.438	3.028	3.929	4.428	5.184	6.467	6.210	6.250	6.783	7.169	7.084	6.657	6.895	7.091	7.666	7.353	7.617	7.983
Gaseous and liquid biofuels ³⁾	272	281	341	343	541	574	728	1.004	869	915	1.127	1.147	1.130	2.348	2.590	2.840	3.099	3.473	3.400	3.679
Biomethane	-	-	-	-	-	-	-	-	524	951	1.490	1.471	1.404	1.960	2.028	2.235	2.401	2.280	2.127	1.963
Geothermal energy, ambient heat, solarthermal energy	-	-	-	-	-	-	-	-	61	87	177	223	180	372	484	552	557	913	866	999
Total	6.892	7.899	8.261	9.744	10.575	11.565	12.640	14.885	16.035	16.428	17.989	18.528	18.791	21.347	22.130	22.346	24.264	24.095	24.305	25.414

Figures in [%]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
for information: share of grid losses and consumption in the transformation sector ⁴⁾	8,9	9,1	8,7	8,9	8,9	8,4	10,0	10,7	10,9	12,6	12,2	12,7	12,6	13,8	11,2	11,2	11,3	12,0	12,2	12,2

Grid losses are not taken into account

- 1) biogenic fraction of waste in waste incineration plants estimated at 50 %, from 2008 only municipal waste
- 2) including sewage sludge
- 3) sum of gaseous and liquid biomasses: biogas, biomethane, sewage gas, landfill gas and liquid biofuels; until 2012 including geothermal energy, near-surface geothermal energy, ambient heat and biomethane; reported separately as of 2013
- 4) share of grid losses including consumption in the transformation sector, with reference to the total district heat production (renewable and fossil energy sources)

Sources: AGEE-Stat based on StBA [26]

Table 5.2: Final energy consumption from renewable sources for heating and cooling by sectors

values before 2005 hidden

Status: September 2025

[Back to directory](#)

Total	Figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	Solid biofuels ¹⁾	83.183	92.935	97.902	110.221	104.669	128.630	116.806	128.244	130.237	110.957	111.723	110.051	112.169	112.526	114.157	109.255	123.602	130.949	112.906	114.702
	Gaseous and liquid biofuels ²⁾	4.207	5.181	8.622	9.104	10.987	13.410	14.231	13.658	12.779	14.115	14.912	15.662	16.446	16.182	16.368	17.522	17.300	17.655	16.827	16.751
	Biomethane	-	-	-	-	-	-	-	-	2.066	2.437	2.661	2.755	2.513	2.172	2.182	2.232	2.853	2.790	3.115	3.044
	Biogenic fraction of waste ³⁾	3.210	3.611	5.743	855	966	1.245	2.086	2.411	4.188	4.188	4.419	4.233	4.807	5.877	6.314	6.513	6.256	6.168	5.241	5.060
	Geothermal energy, ambient heat, solarthermal energy	5.715	6.409	7.044	8.155	9.565	11.023	11.808	13.132	14.174	14.629	16.037	17.092	18.147	19.717	20.405	21.600	23.617	25.329	27.332	29.624
	Total	96.315	108.136	119.311	128.335	126.187	154.307	144.931	157.445	163.444	146.326	149.752	149.793	154.082	156.474	159.426	157.122	173.628	182.891	165.421	169.181

Households	Figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	Solid biofuels ¹⁾	52.222	61.496	64.093	74.020	65.220	79.304	71.904	83.014	85.941	69.483	69.477	66.289	68.393	70.192	71.154	66.874	76.874	85.434	66.186	66.088
	Gaseous and liquid biofuels ²⁾	162	381	948	943	1.421	1.753	2.055	2.466	2.277	2.591	2.822	3.008	3.176	2.994	3.014	3.109	3.212	3.294	3.072	3.124
	Biomethane	-	-	-	-	-	-	-	-	554	644	662	697	659	563	574	577	624	691	804	714
	Solarthermal energy	2.649	3.138	3.504	4.032	4.772	5.090	5.840	6.094	6.184	6.696	7.218	7.265	7.491	8.566	8.293	8.621	8.174	9.306	8.713	8.399
	Geothermal energy ⁴⁾	82	50	47	67	95	164	184	245	253	236	267	346	323	267	294	281	441	414	453	487
	Near-surface geothermal energy, ambient heat ⁴⁾	2.214	2.400	2.637	3.178	3.700	4.682	4.616	5.552	6.431	6.213	7.174	7.973	8.645	8.909	9.798	10.553	12.667	13.179	15.532	17.881
	Total	57.329	67.465	71.229	82.240	75.208	90.993	84.599	97.371	101.640	85.863	87.620	85.578	88.687	91.491	93.127	90.015	101.992	112.318	94.760	96.693

TCS sector	Figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	Solid biofuels ¹⁾	9.695	11.120	11.443	16.046	16.476	21.237	15.813	17.437	18.696	14.945	17.137	16.731	17.450	17.812	19.219	19.101	21.908	19.926	21.429	22.663
	Gaseous biofuels ²⁾	2.514	2.846	4.307	4.341	5.572	6.534	7.452	8.503	7.809	8.645	9.436	9.965	10.514	10.355	10.268	10.477	10.753	10.979	10.370	10.529
	Liquid biofuels, biodiesel in agriculture and forestry	1.110	1.683	2.596	2.591	3.148	2.762	2.137	2.016	1.910	2.066	1.912	1.894	1.935	2.096	2.146	3.030	2.428	2.266	2.349	1.977
	Biomethane	-	-	-	-	-	-	-	-	1.453	1.725	1.807	1.854	1.736	1.444	1.459	1.407	1.529	1.623	1.889	1.614
	Solarthermal energy	208	226	242	261	289	293	319	319	312	327	343	338	341	385	369	381	358	403	375	361
	Geothermal energy ⁴⁾	416	436	440	418	464	462	523	536	554	732	528	579	666	721	647	666	659	669	660	667
	Near-surface geothermal energy, ambient heat ⁴⁾	146	159	174	199	245	331	326	386	440	425	507	591	681	743	875	968	1.177	1.224	1.443	1.661
	Total ⁵⁾	14.441	16.806	19.515	24.092	26.432	31.863	26.589	29.197	31.174	28.865	31.670	31.952	33.323	33.552	34.983	36.030	38.812	37.090	38.515	39.472

Industry	Figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	Solid biofuels ¹⁾	21.266	20.319	22.366	20.155	22.973	28.089	29.089	27.793	25.600	26.529	25.109	27.031	26.326	24.522	23.784	23.280	24.820	25.589	25.291	25.951
	Gaseous and liquid biofuels ²⁾	421	271	771	1.229	846	2.361	2.587	673	783	813	742	795	821	741	940	906	907	1.116	1.036	1.121
	Biomethane	-	-	-	-	-	-	-	-	59	68	192	204	118	165	149	248	700	476	422	716
	Biogenic fraction of waste ³⁾	2.858	3.275	5.430	619	728	1.003	2.067	2.411	4.188	4.188	4.419	4.233	4.807	5.877	6.314	6.513	6.256	6.168	5.241	5.060
	Geothermal energy, ambient heat, solarthermal energy	-	-	-	-	-	-	-	-	-	-	-	-	-	126	129	130	140	134	155	168
	Total	24.545	23.865	28.567	22.003	24.547	31.453	33.743	30.877	30.630	31.598	30.462	32.263	32.072	31.433	31.316	31.077	32.823	33.483	32.345	33.016

1) including sewage sludge

2) sum of gaseous and liquid biomasses: biogas, biomethane, sewage gas, landfill gas and liquid biofuels; since 2013 biomethane depicted separately

3) biogenic fraction of waste in waste incineration plants estimated at 50 %, from 2008 only municipal waste

4) before 2003 balneological plants are not taken into account

5) sum of biogas, biomethane, sewage gas and landfill gas; biomethane since 2013 depicted separately

6) Based on G28 and FH-06, renewable heat from heat pumps (air-water, water-water, brine-water, process water and gas heat pumps)

7) includes low amounts of energy from the biogenic fraction of waste of the TCS sector

Table 5.3: Additional differentiation: Ground source geothermal energy and environmental heat (heat pumps) ¹

values before 2005 hidden

Status: September 2025

[back to directory](#)

Total Thermal Capacity	figures in [1,000 units]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	Electric Heating Heat Pumps	117	141	167	203	236	270	310	353	396	438	481	535	596	659	726	812	920	1,061	1,343	1,492
	of which: Air-to-Water Heat Pumps	17	26	39	57	75	97	126	158	192	225	260	302	350	400	454	525	616	742	1,006	1,145
	of which: Brine-to-Water Heat Pumps	83	96	108	124	137	148	158	167	175	183	190	200	212	224	236	250	266	280	297	306
	of which: Water-to-Water Heat Pumps	17	19	20	22	24	25	26	28	29	30	31	33	34	35	36	37	38	39	40	41
	Domestic Hot Water Heat Pumps	229	214	198	190	182	173	166	163	163	166	170	175	183	192	204	226	244	276	354	391
	Gas-Driven Heat Pumps	0,1	0,1	0,1	0,2	0,3	0,6	1,0	1,5	2,3	3,3	4,8	6,2	7,2	8,2	9,0	9,8	10,5	11,3	12,0	12,0
Total		346	355	365	393	418	444	477	518	561	607	656	716	786	859	939	1,048	1,175	1,348	1,709	1,895

Thermal Work / Total Heat	figures in [MW]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	Electric Heating Heat Pumps	1.629	1.897	2.177	2.549	2.904	3.337	3.813	4.315	4.781	5.204	5.642	6.160	6.748	7.362	7.994	8.821	9.855	11.209	13.944	15.371
	of which: Air-to-Water Heat Pumps	247	360	515	727	938	1.235	1.577	1.959	2.321	2.646	2.995	3.398	3.853	4.329	4.835	5.513	6.383	7.586	10.129	11.462
	of which: Brine-to-Water Heat Pumps	1.090	1.224	1.330	1.465	1.584	1.700	1.815	1.914	1.999	2.081	2.150	2.245	2.357	2.475	2.589	2.725	2.876	3.013	3.186	3.268
	of which: Water-to-Water Heat Pumps	292	313	332	357	382	402	421	442	461	477	497	517	538	558	570	583	596	610	629	641
	Domestic Hot Water Heat Pumps	458	427	395	380	363	345	332	325	325	332	340	350	365	385	409	451	489	552	708	781
	Gas-Driven Heat Pumps	4	6	7	9	16	27	42	63	89	123	172	216	246	276	301	324	347	369	390	388
Total		2.091	2.330	2.579	2.938	3.283	3.709	4.187	4.703	5.195	5.659	6.154	6.726	7.359	8.023	8.704	9.596	10.691	12.130	15.042	16.540

Final Energy Consumption Electricity / Gas	figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	Electric Heating Heat Pumps	3.282	3.553	3.883	4.743	5.583	7.320	6.923	8.485	9.961	9.144	10.750	12.042	13.042	13.415	14.854	15.884	19.613	19.761	23.075	26.418
	of which: Air-to-Water Heat Pumps	471	616	843	1.262	1.725	2.576	2.747	3.717	4.711	4.564	5.601	6.527	7.324	7.771	8.853	9.761	12.475	13.089	16.239	19.449
	of which: Brine-to-Water Heat Pumps	2.214	2.328	2.427	2.794	3.107	3.832	3.385	3.874	4.268	3.726	4.187	4.482	4.652	4.601	4.909	5.032	5.898	5.539	5.698	5.825
	of which: Water-to-Water Heat Pumps	597	609	613	687	751	912	791	894	982	854	962	1.033	1.066	1.043	1.092	1.091	1.240	1.133	1.138	1.144
	Domestic Hot Water Heat Pumps	272	251	233	219	211	202	194	189	188	191	196	203	211	222	235	255	279	310	375	444
	Gas-Driven Heat Pumps	6	10	13	17	26	43	70	107	156	218	303	398	474	535	591	641	689	735	780	803
Total		3.560	3.814	4.129	4.979	5.820	7.565	7.187	8.781	10.305	9.553	11.249	12.643	13.727	14.172	15.680	16.780	20.581	20.808	24.230	27.665

Final Energy Consumption Electricity / Gas	figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	Electric Heating Heat Pumps	1.081	1.144	1.214	1.504	1.775	2.445	2.123	2.697	3.252	2.690	3.281	3.723	3.989	3.940	4.381	4.593	6.018	5.602	6.384	7.162
	of which: Air-to-Water Heat Pumps	207	251	316	479	651	1.030	953	1.352	1.767	1.458	1.880	2.220	2.443	2.443	2.790	2.986	4.107	3.871	4.628	5.388
	of which: Brine-to-Water Heat Pumps	688	707	716	821	904	1.141	947	1.091	1.206	1.001	1.138	1.221	1.257	1.220	1.300	1.319	1.577	1.435	1.461	1.480
	of which: Water-to-Water Heat Pumps	186	186	182	204	220	274	223	254	279	231	263	282	289	277	291	288	334	296	295	294
	Domestic Hot Water Heat Pumps	114	104	95	87	82	77	72	69	67	66	67	68	70	73	76	81	88	96	114	133
	Gas-Driven Heat Pumps	4	6	9	11	17	31	50	78	113	159	220	288	342	384	423	457	490	521	551	567
Total		1.199	1.254	1.318	1.602	1.874	2.553	2.245	2.844	3.432	2.915	3.568	4.079	4.401	4.397	4.880	5.131	6.596	6.219	7.049	7.862

Renewable Heat	figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	Electric Heating Heat Pumps	2.201	2.409	2.669	3.240	3.808	4.875	4.801	5.789	6.708	6.454	7.469	8.320	9.053	9.474	10.473	11.291	13.594	14.159	16.691	19.254
	of which: Air-to-Water Heat Pumps	265	365	527	783	1.074	1.546	1.794	2.365	2.944	3.106	3.721	4.307	4.881	5.328	6.063	6.775	8.368	9.218	11.611	14.060
	of which: Brine-to-Water Heat Pumps	1.525	1.621	1.711	1.973	2.203	2.691	2.439	2.783	3.061	2.725	3.049	3.262	3.395	3.381	3.608	3.713	4.320	4.104	4.237	4.344
	of which: Water-to-Water Heat Pumps	411	423	431	484	531	638	568	641	703	623	699	751	777	765	802	803	906	837	843	850
	Domestic Hot Water Heat Pumps	157	147	138	132	129	125	122	120	121	125	129	135	141	149	159	174	191	214	261	311
	Gas-Driven Heat Pumps	2	3	4	6	8	13	20	29	42	59	83	110	132	151	168	184	199	214	229	236
Total		2.360	2.559	2.811	3.378	3.945	5.013	4.943	5.938	6.871	6.638	7.681	8.565	9.326	9.774	10.800	11.649	13.984	14.587	17.181	19.801

¹ Based on G2B and FH-ISE

Table 6: Final energy consumption from renewable sources in the transport sector ¹

values before 2005 hidden

back to directory

Status: September 2025

Figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Biodiesel including HVO	17.666	27.938	32.282	25.873	22.966	24.359	23.556	24.628	21.945	22.676	20.829	20.896	21.354	22.329	22.109	30.148	25.097	24.818	25.626	21.135
thereof: Biodiesel	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	21.610	21.005	21.507	22.145	18.704
thereof: blended Biodiesel	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	21.559	20.933	21.456	22.083	18.642
thereof: neat Biodiesel	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	51	72	51	62	62
thereof: HVO	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	8.538	4.092	3.310	3.481	2.431
thereof: blended HVO	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	8.514	4.019	3.200	3.066	1.710
thereof: neat HVO	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	24	73	110	415	721
Vegetable oil	1.828	7.206	8.533	4.042	961	574	188	251	0	52	10	31	31	10	21	21	10	10	10	10
Biomethanol	1.780	3.828	3.391	4.608	6.576	8.537	9.031	9.149	8.832	9.002	8.589	8.604	8.464	8.685	8.353	8.014	8.412	8.692	9.142	9.112
Biomethane ²⁾	0	0	0	4	13	75	92	333	483	449	345	379	445	389	660	884	965	1.332	1.954	3.549
thereof: CNG	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	884	965	1.304	1.435	1.507
thereof: LNG	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0	28	519	2.042
RFNBO	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0
thereof: renewable hydrogen	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0
RE electricity consumption in transport ³⁾	1.353	1.484	1.762	1.699	1.925	2.077	2.494	2.862	3.017	3.169	3.523	3.732	4.329	4.581	4.898	5.237	5.366	6.524	8.102	9.171
thereof: road transport	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	302	578	1.137	1.972	3.196
thereof: rail transport	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	4.935	4.788	5.387	6.130	5.975
Total	22.627	40.456	45.968	36.226	32.441	35.622	35.361	37.223	34.277	35.348	33.296	33.642	34.623	35.994	36.041	44.304	39.850	41.376	44.834	42.977

1) consumption of fuels in the transport sector (excluding consumption in agriculture, forestry and military)
 2) Until 2021 exclusively Bio CNG, from 2022 Bio CNG and Bio LNG; based on the net calorific value
 3) calculated on the share of renewables-based electricity generation in gross electricity consumption in each year

Sources: AGEE-Stat based on BAFA [5]; BLE [3]; [4]; BMF [6]; The German Gas and Hydrogen Industry [34]; StBA [25]; UBA [33]; FNR; ZSW; DBFZ; AGQM; UFOP; partially preliminary data

Table 7: Development of energy consumption in Germany

values before 2005 hidden

Status: September 2025

[back to directory](#)

Figures in [TWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Gross electricity consumption ¹⁾	618,3	622,9	624,9	621,5	584,5	618,9	610,4	610,1	607,2	594,5	600,7	600,6	601,9	594,8	577,6	557,9	574,3	551,7	520,9	528,3
Production of district heating	136,1	135,8	130,1	133,0	130,4	143,1	129,8	133,6	135,6	121,8	127,1	130,3	130,5	130,2	127,2	118,3	137,8	119,7	115,4	120,6
Final energy consumption for heating and cooling (including district heat) ²⁾	1.275,3	1.341,3	1.199,7	1.313,5	1.208,6	1.347,0	1.239,6	1.259,1	1.306,8	1.175,1	1.217,7	1.238,3	1.241,3	1.210,1	1.229,0	1.201,0	1.266,9	1.167,6	1.075,7	1.076,8
Final energy consumption for heating and cooling (without electricity and district heat) ³⁾	1.151,5	1.217,7	1.081,0	1.192,3	1.089,8	1.215,9	1.122,8	1.139,9	1.186,2	1.068,7	1.106,1	1.124,6	1.127,3	1.097,6	1.115,6	1.095,1	1.143,9	1.062,3	974,4	970,3
thereof in: Households	538,3	553,8	453,5	542,3	505,8	550,8	468,0	508,8	538,9	448,5	472,0	485,4	480,5	484,7	512,6	505,3	512,2	486,1	441,5	438,2
thereof in: TCS	191,6	230,0	180,7	207,0	195,5	209,4	191,3	188,3	211,8	192,6	199,5	190,5	192,1	169,1	172,6	172,8	177,3	164,6	151,5	149,7
thereof in: Industry	421,6	433,9	446,8	443,0	388,5	455,7	463,5	442,8	435,5	427,6	434,6	448,7	454,7	443,8	430,4	417,0	454,4	411,6	381,4	382,4
Final energy consumption in the transport sector ⁴⁾	634,0	654,9	622,5	610,7	606,2	611,9	617,5	612,4	623,9	630,0	639,8	650,6	662,8	646,6	649,9	591,2	591,9	598,8	588,6	585,3

Figures in [PJ]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Primary energy consumption ⁵⁾	14.506	14.852	14.198	14.381	13.513	14.137	13.523	13.630	13.916	13.244	13.384	13.510	13.528	13.190	12.821	11.891	12.465	11.692	10.633	10.542

1) gross electricity production by fossil fuels accounting to AGEB and AGEE-Stat, status September 2024, data on electricity trade according to StBA

2) sectoral local consumption of energy for application purposes of heating and cooling, without electricity consumption, including district heat; calculation based on Working Group on Energy Balances (AGEB) and AGEE-Stat, status September 2024

3) sectoral local consumption of energy for application purposes of heating and cooling, without secondary energy carriers electricity and district heat; calculation based on Working Group on Energy Balances (AGEB) and AGEE-Stat, status September 2024

4) calculation based on Working Group on Energy Balances (AGEB) and AGEE-Stat, without energy consumption for international aviation, status September 2024

5) calculation based on Working Group on Energy Balances (AGEB) and AGEE-Stat, calculated by the "physical energy content" principle, status September 2024

Sources: AGEE-Stat based on AGEB [1]; StBA [30]; Eurostat [18]; IEA [22]; EP/ER [16], [17]; partially preliminary data

Table 8.1: Greenhouse gas emissions avoided through the use of renewable energy sources ¹

values before 2005 hidden

Status: September 2025

[back to directory](#)

Total	Figures in [1.000 t CO ₂ -eq.]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	by RE gross electricity production	56.052	55.704	66.094	65.196	67.901	74.157	91.539	92.835	95.742	112.945	129.309	129.220	142.161	151.277	177.450	184.226	172.568	184.329	198.873	205.163
	by RE final energy consumption for heating and cooling	20.351	22.314	24.729	26.835	27.318	32.159	30.996	33.292	34.150	31.578	32.845	33.248	33.306	36.108	36.572	36.256	40.076	42.724	39.028	40.007
	by RE final energy consumption in the transport sector ^{1) 2)}	4.248	7.684	8.728	6.913	6.117	6.663	6.540	7.077	6.478	6.746	6.359	6.972	7.397	7.730	7.511	11.353	10.495	11.335	12.986	13.649
	Total avoided GHG-emissions	80.651	85.702	99.551	98.944	101.336	112.979	129.075	133.204	136.370	151.269	168.513	169.440	182.864	195.115	221.533	231.835	223.139	238.388	250.887	258.819

by RE gross electricity production	Figures in [1.000 t CO ₂ -eq.]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	Hydropower	20.794	17.367	17.850	16.140	15.417	16.898	14.855	16.766	16.402	15.584	14.887	15.878	15.022	13.815	16.402	15.192	15.929	14.328	16.072	18.101
	Wind energy onshore	22.487	23.445	30.137	28.812	28.482	27.654	38.006	33.854	36.733	43.563	53.486	49.831	61.683	65.372	77.518	79.970	70.052	75.648	89.355	85.176
	Wind energy offshore	0	0	0	0	28	127	429	493	653	1.138	6.068	9.121	12.544	14.207	19.279	21.236	18.857	19.497	18.424	20.024
	Solar Photovoltaic	602	1.345	1.857	3.050	4.624	8.222	14.320	16.766	18.288	23.618	25.555	25.047	24.958	28.494	31.969	34.997	35.080	42.694	44.338	51.615
	Solid biofuels	6.328	6.827	6.617	6.472	6.987	7.568	7.984	7.499	6.894	7.867	7.869	7.632	7.225	7.746	8.370	8.557	8.070	8.036	7.874	7.621
	Liquid biofuels	82	414	544	613	958	743	229	187	140	189	234	270	226	203	183	147	76	24	25	13
	Biogas	772	1.572	3.949	5.129	6.562	7.664	10.108	11.715	10.942	13.676	14.007	14.116	13.550	14.013	15.886	16.232	16.524	16.428	15.062	15.232
	Biomethane	0	0	10	22	41	192	316	523	717	1.245	1.474	1.476	1.325	1.425	1.641	1.693	1.813	1.703	1.728	1.551
	Sewage gas	1.006	695	676	710	760	802	883	836	760	880	897	926	903	1.001	1.122	1.128	1.118	1.098	1.087	1.086
	Landfill gas	981	718	661	560	529	449	433	341	280	287	256	230	209	196	202	173	173	142	146	120
	Biogenic fraction of waste	3.000	3.321	3.793	3.678	3.502	3.821	3.964	3.841	3.892	4.840	4.498	4.591	4.424	4.701	4.748	4.746	4.715	4.597	4.632	4.480
	Geothermal energy	0	0	0	10	11	17	12	14	41	58	78	102	92	104	130	155	161	134	130	144
	Total	56.052	55.704	66.094	65.196	67.901	74.157	91.539	92.835	95.742	112.945	129.309	129.220	142.161	151.277	177.450	184.226	172.568	184.329	198.873	205.163

by RE final energy consumption for heating and cooling	Figures in [1.000 t CO ₂ -eq.]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	Solid biofuels (households)	7.460	8.831	9.380	11.031	9.837	11.864	10.835	12.674	13.303	11.006	10.733	10.378	10.675	11.528	11.399	10.699	12.616	14.206	10.946	10.848
	Solid biofuels (TCS sector)	2.451	2.847	2.941	4.085	4.177	5.319	4.039	4.479	4.847	3.826	4.271	4.185	3.720	4.018	4.215	4.120	4.827	4.542	4.715	4.950
	Solid biofuels (industry)	6.228	5.860	6.460	6.171	6.991	7.938	8.404	7.860	6.953	7.251	7.213	7.743	7.386	7.409	6.954	6.692	7.131	7.706	7.197	7.321
	Solid biofuels (HP/CHP)	498	484	544	675	887	981	1.138	1.420	1.203	1.205	1.397	1.528	1.480	1.414	1.518	1.577	1.693	1.693	1.752	1.835
	Liquid biofuels	269	406	641	716	800	720	533	456	446	487	480	509	549	577	576	716	561	536	498	456
	Biogas	80	136	362	358	513	781	990	868	928	1.050	1.568	1.692	1.878	2.323	2.304	2.322	2.566	2.794	2.531	2.603
	Biomethane	0	0	3	10	20	76	118	187	365	471	883	815	782	548	618	646	738	735	712	691
	Sewage gas	505	446	444	475	468	478	502	464	412	411	510	522	553	664	673	697	732	728	716	723
	Landfill gas	70	69	64	47	47	36	32	28	27	29	41	41	38	41	40	35	29	25	20	19
	Gaseous biofuels	1.649	1.926	2.454	1.528	1.513	1.644	1.857	2.074	2.665	2.611	2.471	2.352	2.475	3.061	3.451	3.556	3.707	3.730	3.504	3.567
	Solarthermal energy	791	927	1.028	1.184	1.403	1.534	1.757	1.829	1.854	2.004	1.981	1.986	2.045	2.481	2.390	2.478	2.353	2.722	2.482	2.393
	Geothermal energy	157	155	154	164	188	202	213	240	256	314	278	322	327	380	399	410	449	542	534	572
	Near-surface geothermal energy, ambient heat	193	227	254	391	474	586	578	713	891	913	1.019	1.175	1.398	1.664	2.035	2.308	2.674	2.765	3.421	4.029
	Total	20.351	22.314	24.729	26.835	27.318	32.159	30.996	33.292	34.150	31.578	32.845	33.248	33.306	36.108	36.572	36.256	40.076	42.724	39.028	40.007

by RE final energy consumption in the transport sector ^{3) 4)}	Figures in [1.000 t CO ₂ -eq.]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
	Biodiesel incl. HVO	3.480	5.462	6.357	5.123	4.490	4.669	4.594	5.065	4.516	4.731	4.432	4.904	5.183	5.420	5.193	8.306	7.083	7.212	7.481	6.170
	Vegetable oil	374	1.366	1.598	761	189	115	34	45	0	9	2	6	6	2	4	5	2	2	2	2
	Bioethanol	394	856	773	1.028	1.435	1.846	1.895	1.902	1.867	1.915	1.855	1.978	2.108	2.222	2.187	2.477	2.549	2.626	2.785	2.776
	Biomethane	0	0	0	1	3	33	17	65	95	91	70	84	100	86	127	271	306	575	1.122	2.038
	RE electricity consumption in transport ⁴⁾	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	294	555	920	1.596	2.663
	Total	4.248	7.684	8.728	6.913	6.117	6.663	6.540	7.077	6.478	6.746	6.359	6.972	7.397	7.730	7.511	11.353	10.495	11.335	12.986	13.649

1) not including biogenic CO2 from the thermal use of wood based solid biofuels
2) consumption of biogenic fuels in the transport sector (excluding consumption in agriculture, forestry and military and excluding electricity consumption in the transport sector)
3) biofuels based on data of the Federal Office for Agriculture and Food (BLE) for the year 2023 and the fossil reference values according § 3 and § 10 of 38. BImSchV
4) Accounting of emission reductions from the use of electricity in road transport, taking into account the current electricity mix

Source: UBA [30] based on sources quoted there; partially preliminary data

Table 8.2: Acidifying emissions avoided through the use of renewable energy sources ²

values before 2005 hidden
[back to directory](#)

Status: September 2025

Figures in [1.000 t SO ₂ -eq.]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
by RE gross electricity production	33,6	25,9	23,9	23,9	21,6	23,3	32,5	21,5	19,5	30,0	97,2	82,2	99,1	106,7	124,0	122,3	114,6	129,8	150,5	156,0
by RE final energy consumption for heating and cooling	-1,6	-2,8	-4,2	0,9	-2,0	-5,2	-5,5	-8,0	-11,8	-9,8	-13,2	-14,0	-21,2	-21,4	-22,5	-26,2	-27,6	-20,8	-21,3	-21,0
by RE final energy consumption in the transport sector ¹⁾	-1,2	-3,6	-11,9	-7,8	-4,8	-7,0	-4,6	-6,0	-6,9	-4,8	-3,7	-3,6	-19,0	-18,6	-17,4	-7,0	-5,6	-3,9	-1,6	-0,9
Total avoided acidifying agents ²⁾	30,8	19,5	7,8	17,0	14,8	11,1	22,4	7,5	0,8	15,4	80,3	64,6	58,9	66,7	84,1	89,1	81,4	105,1	127,6	134,1

1) consumption of biogenic fuels in the transport sector (excluding consumption in agriculture, forestry and military and excluding electricity consumption in the transport sector)
2) as acidifying emissions only SO₂ and Nox are taking into account

Source: UBA [30] based on sources quoted there; partially preliminary data

Table 8.3: Avoidance factors of renewable energy sources 1990 to 2024 ²

values before 2005 hidden

Status: September 2025

[back to directory](#)

Figures in [g CO ₂ -eq./kWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Avoidance factor of RE gross electricity production	882	765	734	686	700	697	729	640	623	690	680	676	653	671	728	727	720	721	722	718
Avoidance factor of RE final energy consumption for heating and cooling	202	196	198	198	204	198	201	198	195	199	201	203	198	209	207	207	207	210	210	210
Avoidance factor of RE final energy consumption in the transport sector ^{1) 2)}	200	197	197	200	200	199	199	206	207	210	214	233	244	246	241	283	288	299	310	325

Figures in [g CO ₂ -eq./kWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Avoidance factor of RE gross electricity production	0,53	0,36	0,27	0,25	0,22	0,22	0,26	0,15	0,13	0,18	0,51	0,43	0,46	0,47	0,51	0,48	0,48	0,51	0,55	0,55
Avoidance factor of RE final energy consumption for heating and cooling	-0,02	-0,03	-0,03	0,01	-0,01	-0,03	-0,04	-0,05	-0,07	-0,06	-0,08	-0,09	-0,13	-0,12	-0,13	-0,15	-0,14	-0,10	-0,11	-0,11
Avoidance factor of RE final energy consumption in the transport sector ¹⁾	-0,06	-0,09	-0,27	-0,22	-0,16	-0,21	-0,14	-0,17	-0,22	-0,15	-0,12	-0,12	-0,63	-0,59	-0,56	-0,18	-0,16	-0,11	-0,04	-0,03

1) consumption of biogenic fuels in the transport sector (excluding consumption in agriculture, forestry and military and excluding electricity consumption in the transport sector)

2) based on data of the Federal Office for Agriculture and Food (BLE) for the year 2023 and the fossil reference values according § 3 and § 10 of 38. BImSchV

3) as acidifying emissions only SO₂ and NO_x are taking into account

Source: UBA [30] based on sources quoted there; partially preliminary data

Table 9.1: Investments in construction of renewable energy plants

values before 2005 hidden

Status: September 2025

[back to directory](#)

Figures in [million euro] ¹⁾	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Hydropower	240	220	330	370	500	350	300	200	130	90	80	60	60	120	110	100	70	70	10	3
Wind energy onshore	2.490	3.220	2.470	2.540	2.800	2.110	2.860	3.550	4.490	7.060	5.370	6.910	7.450	3.390	1.650	2.190	2.990	3.830	5.960	5.470
Wind energy offshore	-	-	30	170	470	450	610	2.440	4.270	3.940	3.680	3.370	3.400	4.100	2.130	80	280	1.510	1.880	3.580
Solar Photovoltaic	4.840	4.010	5.330	7.970	13.570	19.580	15.860	11.980	3.380	1.450	1.480	1.570	1.660	2.580	3.370	4.920	5.230	8.210	18.840	16.700
Solar thermal energy	630	990	760	1.700	1.490	990	1.060	950	860	790	800	700	540	490	440	530	550	690	420	240
Geothermal energy, ambient heat	410	940	920	1.230	1.140	960	990	1.060	1.090	1.080	1.010	1.210	1.320	1.520	1.410	1.920	2.530	4.580	8.820	5.320
Biomass electricity	1.910	2.270	2.280	1.980	2.020	2.240	3.120	790	700	670	220	270	280	390	350	320	250	240	250	240
Biomass heat	1.510	2.300	1.500	1.760	1.610	1.210	1.320	1.500	1.560	1.320	1.290	1.230	1.230	1.240	1.260	1.940	2.730	3.650	2.770	1.770
Total	12.030	13.950	13.620	17.720	23.600	27.890	26.120	22.470	16.480	16.400	13.930	15.320	15.940	13.830	10.720	12.000	14.630	22.780	38.950	33.323

1) differences in the totals may occur due to roundings

Source: ZSW [32]; partially preliminary data

Table 9.2: Economic stimuli from the operation of renewable energy plants

values before 2005 hidden

Status: September 2025

[back to directory](#)

Figures in [million euro] ¹⁾	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Hydropower	130	130	140	150	160	170	190	190	200	200	200	210	210	210	220	230	230	230	240	240
Wind energy onshore	550	630	710	790	870	970	1.060	1.200	1.360	1.550	1.730	1.890	2.080	2.230	2.300	2.310	2.310	2.300	2.280	2.320
Wind energy offshore	-	-	-	-	10	20	30	60	130	210	280	350	420	500	560	600	620	650	760	940
Solar Photovoltaic	130	190	260	360	530	770	1.040	1.250	1.360	1.400	1.420	1.440	1.470	1.500	1.540	1.600	1.670	1.770	1.960	2.210
Solar thermal energy	50	70	90	110	140	170	190	210	230	240	260	270	290	300	310	320	330	340	350	360
Geothermal energy, ambient heat	240	290	360	440	530	620	730	820	900	980	1.060	1.140	1.230	1.340	1.450	1.580	1.750	2.010	2.410	2.780
Biomass electricity	720	1.100	1.680	2.020	2.420	2.880	3.320	4.080	4.200	4.500	4.650	4.640	4.670	4.670	4.780	4.830	4.650	4.850	4.890	4.830
Biomass heat	1.460	1.740	1.960	2.150	2.450	2.880	2.870	3.120	3.320	3.030	3.180	3.360	3.390	3.340	3.350	3.370	3.830	5.120	5.250	4.780
Biomass fuels	1.790	3.150	3.750	3.530	2.390	2.920	3.690	3.720	3.050	2.640	2.440	2.560	2.710	2.700	2.830	3.540	4.970	6.750	5.150	4.300
Total	5.070	7.300	8.950	9.550	9.500	11.400	13.120	14.650	14.750	14.750	15.220	15.860	16.470	16.790	17.340	18.380	20.360	24.020	23.290	22.760

1) differences in the totals may occur due to roundings

Source: ZSW [32]; partially preliminary data

[back to directory](#)

		RE 2023	Share of renewable energy	avoided GHG-emissions
		[GWh]	[%]	[1.000 t CO ₂ -eq.]
Gross electricity production	Hydropower	19.895	3,8	16.072
	Wind energy onshore	117.877	22,6	89.355
	Wind energy offshore	23.887	4,6	18.424
	Solar Photovoltaic	64.300	12,3	44.338
	Solid biofuels & sewage sludge	10.460	2,0	7.874
	Liquid biofuels	110	< 0,1	25
	Biogas	28.272	5,4	15.062
	Biomethane	3.120	0,6	1.728
	Sewage gas	1.529	0,3	1.087
	Landfill gas	211	< 0,1	146
	Biogenic fraction of waste	5.701	1,1	4.632
	Geothermal energy	195	< 0,1	130
	Total	275.557	52,9	198.873
Final energy consumption for heating and cooling	Solid biofuels & charcoal (households)	66.187	6,2	10.946
	Solid biofuels & charcoal (TCS sector)	21.429	2,0	4.715
	Solid biofuels & sewage sludge (industry)	25.291	2,4	7.197
	Solid biofuels & sewage sludge (HP/CHP)	6.685	0,6	1.752
	Liquid biofuels	2.532	0,2	498
	Biogas	14.906	1,4	2.531
	Biomethane	4.982	0,5	712
	Sewage gas	2.311	0,2	716
	Landfill gas	65	< 0,1	20
	Biogenic fraction of waste	14.278	1,3	3.504
	Solar thermal energy	9.121	0,8	2.482
	Deep geothermal energy	1.796	0,2	534
	Near-surface geothermal energy & ambient heat	17.174	1,6	3.421
	Total	186.757	17,4	39.028
Final energy consumption transport	Biodiesel incl. HVO	25.626	4,4	7.481
	Vegetable oil	10	< 0,1	2
	Bioethanol	9.142	1,6	2.785
	Biomethane	1.954	0,3	1.122
	RE electricity consumption transport	8.102	1,4	1.596
	Total	44.834	7,6	12.986

[back to directory](#)

		RE 2024	Share of renewable energy	avoided GHG-emissions
		[GWh]	[%]	[1.000 t CO ₂ -eq.]
Gross electricity production	Hydropower	22.477	4,3	18.101
	Wind energy onshore	112.846	21,4	85.176
	Wind energy offshore	26.068	4,9	20.024
	Solar Photovoltaic	75.379	14,3	51.615
	Solid biofuels & sewage sludge	10.147	1,9	7.621
	Liquid biofuels	96	< 0,1	13
	Biogas	28.641	5,4	15.232
	Biomethane	2.796	0,5	1.551
	Sewage gas	1.529	0,3	1.086
	Landfill gas	177	< 0,1	120
	Biogenic fraction of waste	5.533	1,0	4.480
	Geothermal energy	215	< 0,1	144
	Total	285.904	54,1	205.163
Final energy consumption for heating and cooling	Solid biofuels & charcoal (households)	66.088	6,1	10.848
	Solid biofuels & charcoal (TCS sector)	22.663	2,1	4.950
	Solid biofuels & sewage sludge (industry)	25.950	2,4	7.322
	Solid biofuels & sewage sludge (HP/CHP)	7.010	0,7	1.835
	Liquid biofuels	2.166	0,2	456
	Biogas	15.432	1,4	2.603
	Biomethane	4.768	0,4	691
	Sewage gas	2.330	0,2	723
	Landfill gas	62	< 0,1	19
	Biogenic fraction of waste	14.535	1,3	3.567
	Solar thermal energy	8.795	0,8	2.393
	Deep geothermal energy	1.915	0,2	572
	Near-surface geothermal energy & ambient heat	19.791	1,8	4.029
	Total	191.505	17,8	40.008
Final energy consumption transport	Biodiesel incl. HVO	21.135	3,6	6.170
	Vegetable oil	10	< 0,1	2
	Bioethanol	9.112	1,6	2.776
	Biomethane	3.549	0,6	2.038
	RE electricity consumption transport	9.171	1,6	2.663
	Total	42.977	7,3	13.649

[back to directory](#)

Methodological changes

National and international reporting obligations increase the demand of reliable and long-term updateable statistical data on the development of all renewable energy sources in Germany. The Working Group on Renewable Energy Statistics (AGEE-Stat) continuously improves and unifies data base and calculation methods. If new information allow conclusions to be drawn on historic values, these data are complemented or corrected. An overview of the implemented methodological changes in comparison to the previous publication is provided below:

Revision of Time Series on Shallow Geothermal Energy and Ambient Heat (Heat Pumps)

Based on official census and microcensus survey results on the current stock of residential buildings heated with heat pumps, the AGEE-Stat model for estimating the stock of heat pumps has been fundamentally revised. In addition to sales statistics from the German Heat Pump Association (Bundesverband Wärmepumpen e.V.) and official census data, information from the official new construction statistics was also incorporated. Furthermore, the model has been expanded to include the influence of weather conditions, which affect both the modeled heat demand and the efficiency of the existing stock of heat pumps in the reporting year.

Inclusion of Plug-in Solar Systems and Differentiated Classification of the Total Photovoltaic System Stock by Size Categories

In the latest update of the AGEE-Stat time series, a more detailed assessment of the installed base of photovoltaic (PV) systems was carried out using data from the Federal Network Agency's Market Master Data Register (Marktstammdatenregister). Additionally, based on a completed expert study, extrapolated data on the total stock, installed capacity, and electricity generation of "plug-in solar systems" (so-called balcony power plants), which had previously not been adequately captured in official statistics, were integrated. For the first time, the PV system portfolio is also broken down by both gross and net capacity across various size categories and by building-integrated and ground-mounted photovoltaic systems (the latter including other types of PV installations).

Further Differentiation of Final Energy Consumption in the Transport Sector

With the recent update of the AGEE-Stat time series, the final energy consumption in the transport sector is now presented in more detail than before. Starting in 2020, biodiesel and HVO are reported separately and, for each year, further distinguished between blended and pure fuels. In the case of biomethane, a differentiation between CNG and LNG has also been introduced from 2020 onward.

Moreover, the consumption of renewable electricity in transport is no longer reported as a single total starting in 2020, but is now broken down into road and rail transport. In addition, the AGEE-Stat time series has been expanded to include "RFNBO / renewable electricity-based hydrogen (EE-hydrogen).

[back to directory](#)

Conversion factors

Joule	J	for energy, work, heat
Watt	W	for capacity, energy flow, heat flow
1 Joule (J) = 1 Newton metre (Nm) = 1 Watt second (Ws)		

Legally binding units in Germany since 1978.

Calorie and derived units such as coal equivalent and oil equivalent are still used as alternatives.

<i>referred to net calorific value</i>		PJ	TWh	Mtce	Mtoe
1 Petajoule	PJ	1	0,2778	0,0341	0,0239
1 Terawatt hour	TWh	3,6	1	0,123	0,0861
1 million tonne of hard coal unit	Mtce	29,308	8,14	1	0,7
1 million tonne of oil equivalent	Mtoe	41,869	11,63	1,429	1

1 TWh = 1 billion kWh	Kilo (k)	10 ³	Tera (T)	10 ¹²
1 GWh = 1 million kWh	Mega (M)	10 ⁶	Peta (P)	10 ¹⁵
1 MWh = 1.000 kWh	Giga (G)	10 ⁹	Exa (E)	10 ¹⁸

Abbreviations

eq.	Equivalent
RE	Renewable energies
EEG	Renewable Energy Sources Act (Erneuerbare-Energien-Gesetz)
FEC	Final energy consumption
N/A	Not quantified
PEC	Primary energy consumption
HP	Heating plant
CHP	Combined heat and power plant

[back to directory](#)

INSTITUTIONS

AGEB	Working Group on Energy Balances e.V., Berlin.
BAFA	Federal Office for Economic Affairs and Export Control, Eschborn.
BDEW	German Association of Energy and Water Industries e.V., Berlin.
BLE	Federal Office of Agriculture and Food, Bonn.
BMLEH	Federal Ministry of Agriculture, Food and Regional Identity, Bonn.
BMUKN	Federal Ministry for the Environment, Climate Action, Nature Conservation and Nuclear Safety, Bonn.
BMWE	Federal Ministry for Economic Affairs and Energy, Berlin.
BNetzA	Federal Network Agency, Bonn.
BSW	German Solar Association, Berlin.
BWE	German Wind Energy Association e.V., Berlin.
BWP	German Heat Pump Association e.V., Berlin.
DBFZ	German Biomass Research Centre, Leipzig.
DENA	German Energy Agency, Berlin.
DEPV	German Energy Pellet Association e.V., Berlin.
DIW	German Institute for Economic Research, Berlin.
FNR	Specialist agency renewable raw materials e.V., Gülzow.
GtV	Geothermal Energy Association e.V., Berlin.
GZB	International Geothermal Center, Bochum
LIAG	Leibniz Institute for Applied Geophysics: Geothermal Information System for Germany, www.geotis.de .
RWI	Leibniz Institute for Economic Research, Essen.
StBA	Federal Statistical Office, Wiesbaden.
TI	Institute of International Forestry and Forest Economics, Hamburg.
UBA	Federal Environment Agency, Dessau-Roßlau.
UL	UL International GmbH, DEWI, Wilhelmshaven.
ÜNB	Information platform of the German Transmission System Operators, www.netztransparenz.de
ZSW	Centre for Solar Energy and Hydrogen Research Baden-Württemberg, Stuttgart.

REFERENCES & LITERATURE

- [1] Working Group on Energy Balances (AGEB): Energy balance of the Federal Republic of Germany, Berlin
- [2] Born, H. et al.: Analysis of the German heat pump market – overview and trends, 2. update, International Geothermal Centre, Bochum, 2017.
- [3] Federal Office of Agriculture and Food (BLE): Evaluation and Progress Reports - Biomass Energy Sustainability Ordinance, Biofuel Sustainability Ordinance, Bonn, https://www.ble.de/EN/Topics/Climate-Energy/Sustainable-Biomass-Production/Information-Materials/information-materials_node.html
- [4] Federal Office of Agriculture and Food (BLE): Applications on the rate for the reduction of greenhouse gas emissions - final data for 2017.
- [5] Federal Office for Economic Affairs and Export Control (BAFA), Federal mineral oil statistik, www.bafa.de.
- [6] Federal Ministry of Finance (BMF) and Office for Biofuel Quotas: Statistical data on fulfilling the biofuel quota, Berlin.

[back to directory](#)

- [7] Federal Ministry for Economic Affairs and Energy (BMWi): Energiedata – national und international developement, Berlin.
- [8] Federal Network Agency (BNetzA): Renewable Energy Sources Act – statistics report on annual accounts, Renewable Energy Sources Act in figures and evaluation of BNetzA register data, Bonn, www.bundesnetzagentur.de
- [9] Federal Government (BReg): Report of the German Federal Government regarding the development of the reduction in greenhouse gas emissions of biofuels, the biomass potential and the amounts of biomethane in the transport fuel market, State: 10 May 2012, <http://dip21.bundestag.de/dip21/btd/17/096/1709621.pdf>
- [10] Federal Government (BReg): National Reports for the implementation of the directive 2003/30/EG from 08 May 2003 on the promotion of the use of biofuels or other renewable fuels in the transport sector.
- [11] Federal Government (BReg): Ordinance on requirements for the sustainable production of biofuels (Biokraftstoff-Nachhaltigkeitsverordnung) of 30 September 2009 (Federal Law Gazette I p. 3182).
- [12] Federal Government (BReg): Ordinance on requirements for the sustainable production of liquid biomass for the power generation (Biomass Energy Sustainability Ordinance – BioSt-NachV) from 23 July 2009 (Federal Law Gazette I p. 2174).
- [13] German Biomass Research Centre (DBFZ) in cooperation with Fraunhofer Institute for Wind Energy Systems (IWES), Private Institute for Sustainable Land Management (INL) und Bosch & Partner GmbH: Power generation from biomass – intermediate and final reports, Leipzig.
- [14] German Wind Energy Institute (UL International (DEWI)): Wind energy in Germany – installation figures for the years 1990 to 2002, www.dewi.de and publications in the DEWI magazine
- [15] Döring, P.; Glasenapp, S.; Mantau, U.: Resource monitoring for wood: Use of fuelwood in private households 2014 - Market volume and used timber assortments, Final report February 2016, Universität Hamburg.
- [16] EP/EC: Directive 2009/28/EC of the European parliament and of the Council of 23 April 2009 on the promotion of the use of energy from renewable sources and amending and subsequently repealing Directives 2001/77/EC and 2003/30/EC. Official Journal of the European Union, L 140/16 of 15 June 2009
- [17] EP/EC: Directive 2015/1513 of the European parliament and of the Council of 9 September 2015 amending Directive 98/70/EC relating to the quality of petrol and diesel fuels and amending Directive 2009/28/EC on the promotion of the use of energy from renewable sources. Official Journal of the European Union, L 239/1 of 15 September 2015
- [18] Eurostat, Statistical Office of the European Communities, Luxembourg: SHARES - Short Assessment of Renewable Energy Sources; <http://ec.europa.eu/eurostat/web/energy/data/shares>
- [19] Agency for Renewable Resources (FNR): Resource monitoring for wood. <https://forst.fnr.de/projekte-und-foerderung/projekte/holzbereitstellung/rohstoffmonitoring-holz/>
- [20] Grawe, J.; Wagner, E.: Use of renewable energy in the electricity industry 1992, 1994, 1996, 1999; in: ew (Elektrizitätswirtschaft).
- [21] Institute for Energetic and Environment (IE), Leipzig, Fichtner GmbH & Co. KG, Stuttgart, Thuringian State Institute for Agriculture (TLL), Jena: Monitoring report on the impact of the Renewable Energy Sources Act on electricity generation from biomass.
- [22] International Energy Agency (IEA), Statistische Amt der Europäischen Gemeinschaften Eurostat, United Nations/ Economic Commission for Europe (UNECE): Energy Questionnaires for Germany.

[back to directory](#)

- [23] Jochem D., Weimar H., Bösch M. et al. (2015) Estimation of wood removals and fellings in Germany: a calculation approach based on the amount of used roundwood. Eur J Forest Res 134(5):869-888, DOI:10.1007/s10342-015-0896-9
- [24] Reinholz, T.; Völler, K.: Brief study – Data of the biomethane market - Gathering and analysis of currently available data as well as retroactive time series, Berlin, July 2018.
- [25] Federal Statistical Office (StBA) : Energy tax statistics – special series 14 series 9.3
- [26] Federal Statistical Office (StBA) : Environment – Waste disposal. Special series 19 series 1.
<https://www.destatis.de/DE/Publikationen/Thematisch/UmweltstatistischeErhebungen/Abfallwirtschaft/Abfallentsorgung.html>
- [27] Federal Statistical Office (StBA) : Statistical surveys in the sectors electricity production and capacity: 066K, 067, 070 and 073.
- [28] Federal Statistical Office (StBA) : Statistical surveys in the heat sector: 060, 062, 064, 066K, 067, 073 and external trade statistics.
- [29] Thünen-Institute of International Forestry and Forest Economics: Fellings and use of roundwood.
<https://www.thuenen.de/en/wf/figures-facts/production-and-use/fellings-and-roundwood-use/>
- [30] Federal Environment Agency (UBA): Emission balance of renewable energy sources. Determining the volume of emissions eliminated in 2018, Dessau-Roßlau, 2019.
- [31] Weimar, H.: From empirical studies to bioenergy statistics: bridging the GAP of unrecorded wood-bioenergy in Germany. In: 9th International Scientific Conference: The path forward for wood products: A global perspective; Baton Rouge, Louisiana, USA, October 5th - 8th 2016. WoodEMA, pp 135-142.
- [32] Centre for Solar Energy and Hydrogen Research Baden-Württemberg (ZSW): Economic impacts of the expansion of renewable energies. Data about renewable energy as an economic factor. Stuttgart, September 2025.
- [33] UBA: Estimate of HVO100, B100, and Vegetable Oil Sales in 2024
- [34] The German Gas and Hydrogen Industry: Sales of Bio-LNG through Member Companies' Filling Stations

EDITING AND PROCESSING

Federal Environment Agency (UBA),
Section V 1.8 - Monitoring of Renewable Energies, Coordination office of the Working Group on Renewable Energy Statistics (AGEE-Stat),
Dessau-Roßlau